Container-based High-Performance Computing Approaches for Simulating Large-Scale Biological Neural Networks

Oliver James and **Sungho Hong**†
Center for Cognition and Sociality, Institute for Basic Science; † sunghohong@ibs.re.kr

Introduction

1. Modeling **large-scale, biophysically detailed neural networks** increasingly relies on **high-performance computing (HPC)**.

2. However, deploying simulations of such networks across diverse HPC environments remains challenging due to **software incompatibilities** and **complex dependencies**.

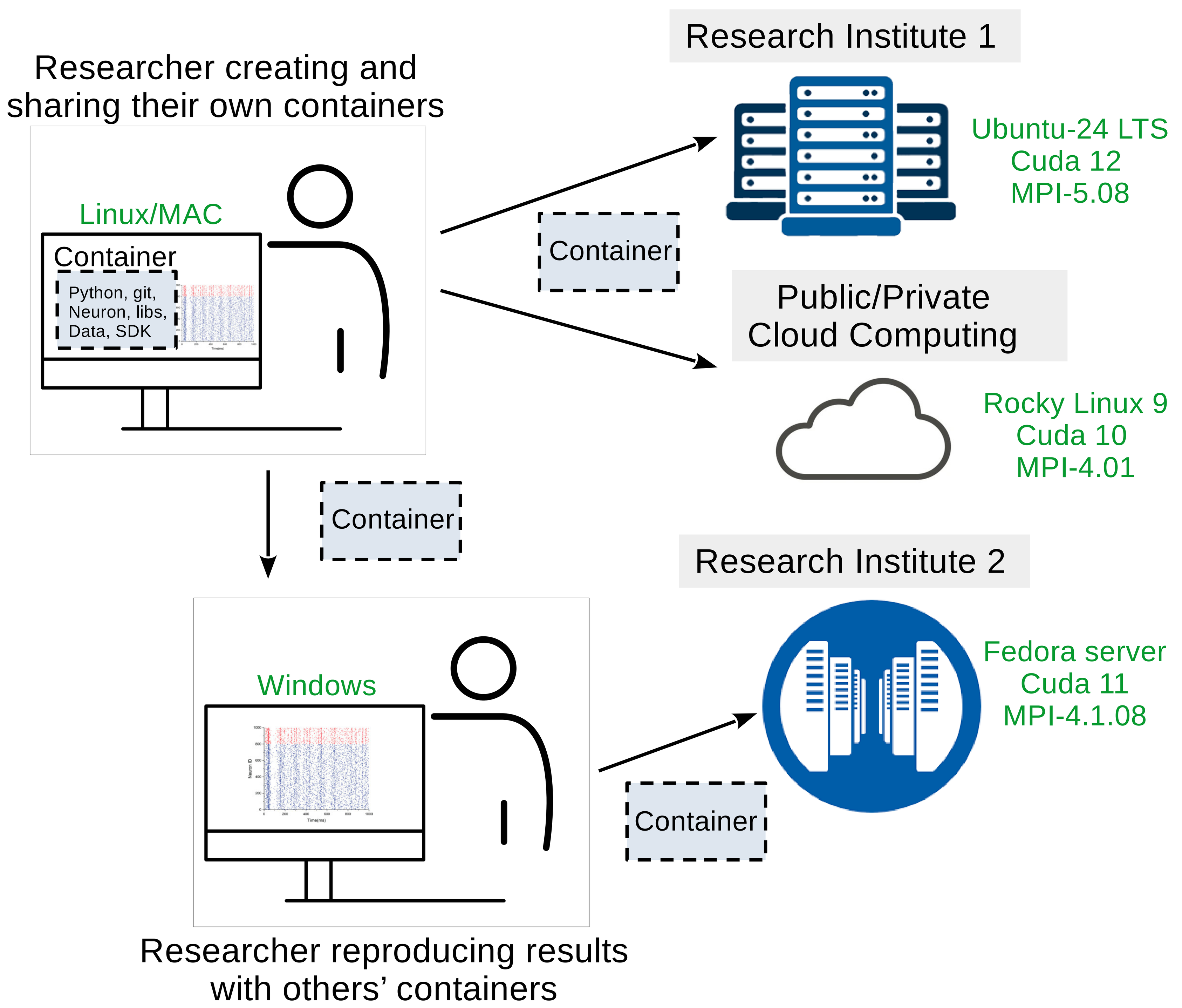
3. We address this issue using **Singularity containers** that encapsulats the entire simulation environment, enabling neuroscientists to achieve portability and reproducibility without requiring root privileges on HPC/Cloud systems.

4. We give practical demonstrations by benchmarking HPC environments with **cerebellar mossy fiber—granule cell network** simulations.

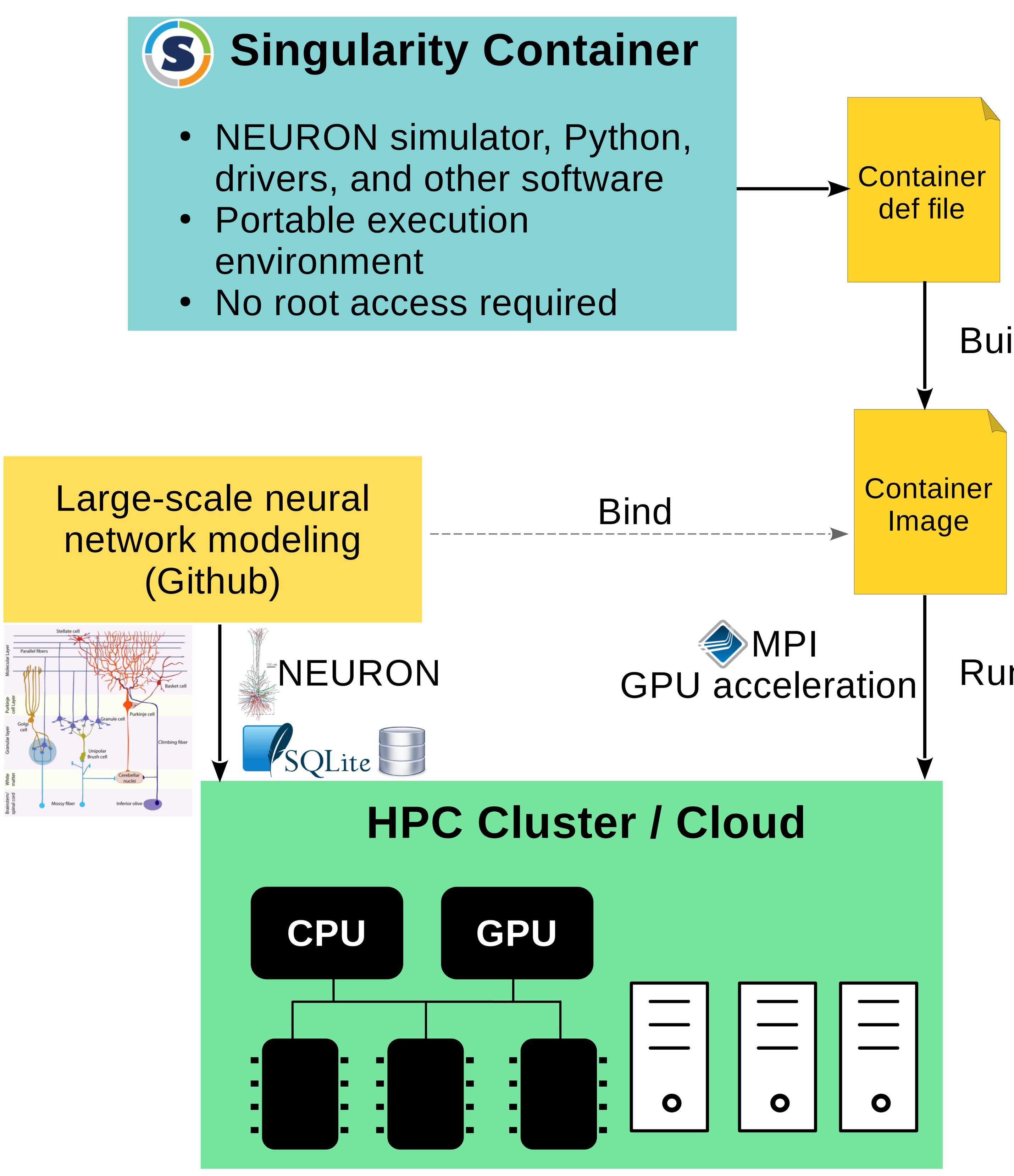
Containerization for Scientific Computing

Container is a lightweight, standalone, and executable package.

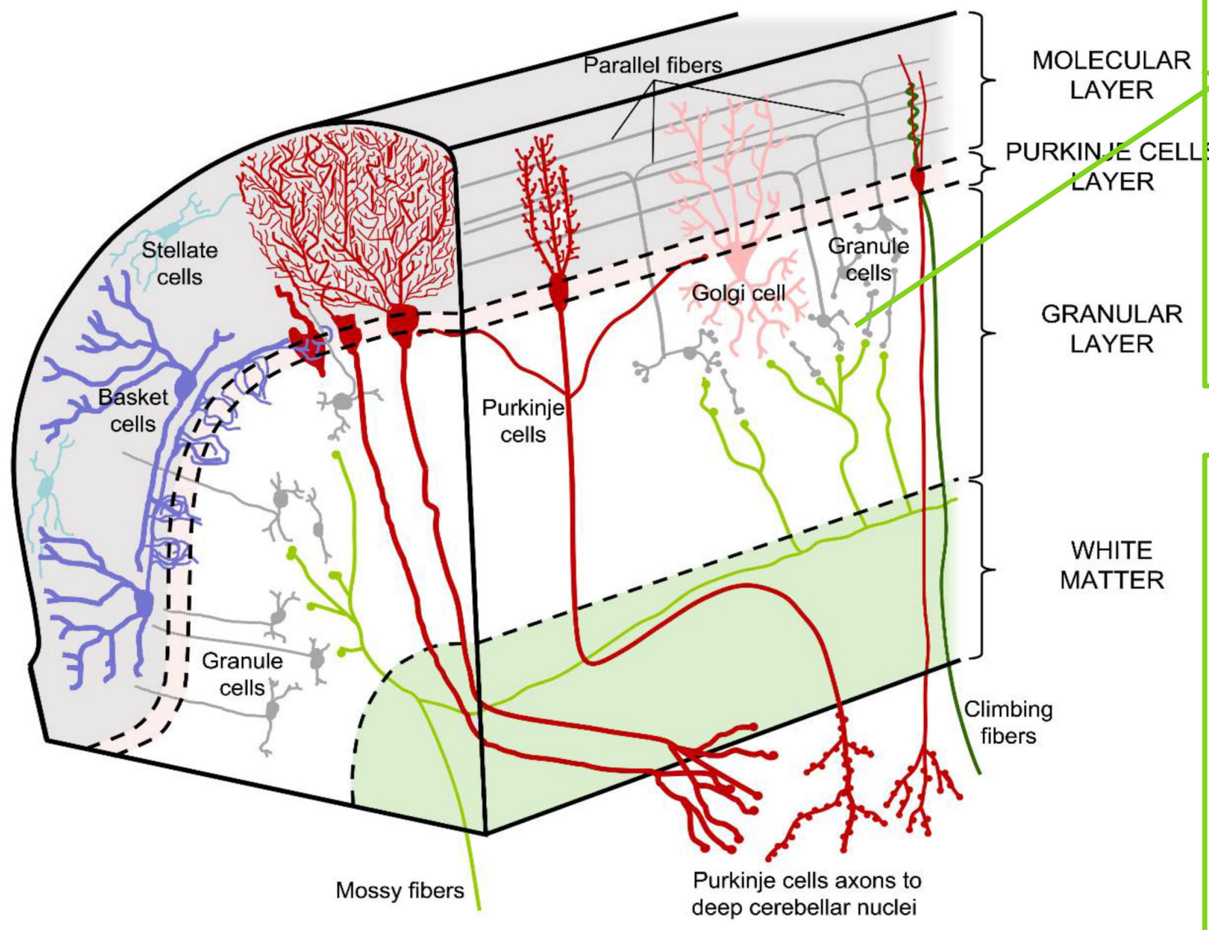
It is designded to include everything needed to run a scientific software.



Our Computational Framework



Benchmarking Data and Network: Cerebellar Mossy Fiber → Granule Cell



EM Data

MF - GRC Connectivity

Nguyen, T.M., Thomas, L.A., Rhoades, J.L. et al. Structured cerebellar connectivity supports resilient pattern separation. **Nature** 613, 543–549 (2023)

MF → 1042 cells
GRC → 3953 cells

Cell Models

MF → spike generators

GRC → Bio-physicaly detailed model with 12 ion-channel mechanisms

Sudhakar, S.K., Hong, S., et.al., 2017. Spatiotemporal network coding of physiological mossy fiber inputs by cerebellar granular layer. **PLOS Compu. Biol** 13

We run multiple copies ($n = 80$) of the network connectivity model simultaneously to simulate extreme computational workloads.

This allows us to rigorously assess the scalability and performance limits of GPUs under high-demand conditions.

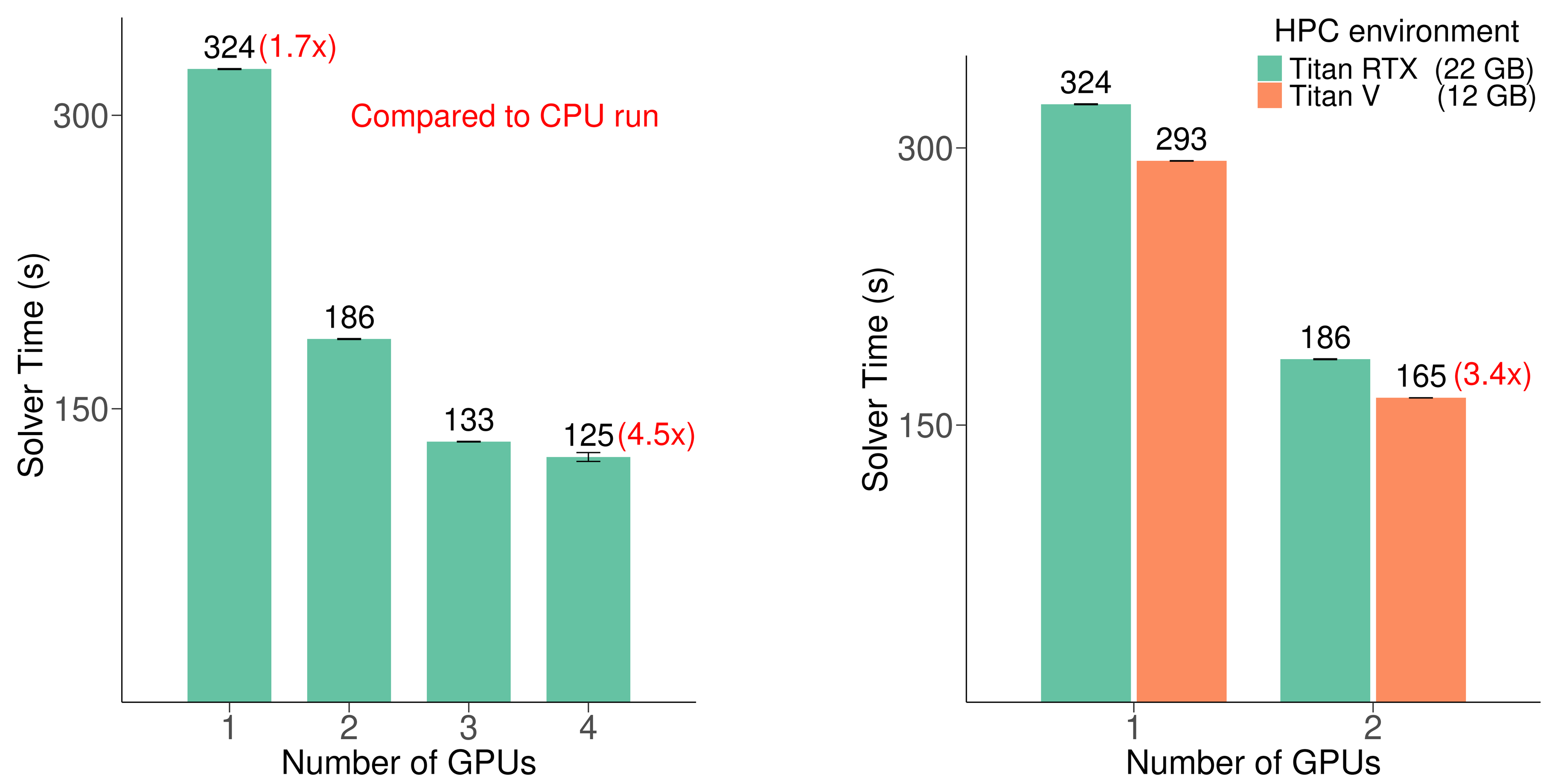
Table: Network Configuration and Simulation Parameters

Number of MF	85,600
Number of GRC	314,000
Number of compartments	942,000
Number of synapses	990,960
Simulation time	2 s
Simulation software	COREneuron

Benchmarking HPC Environments

Table: Performance in CPU parallel computing (30-task MPI)

CPU	Core	Solver Time (s)	Simulator
Intel(R)-Xeon	30	564	CORENeuron-CPU



(a) TITAN RTX architecture: Comparison of solver time for varying numbers of GPUs in COREneuron-GPU simulator

(b) Comparison of solver time between HPC environments

Figure: Solver time comparisons in COREneuron simulations across HPC platforms.

TITAN RTXs outperform 30-core CPUs by up to 4.5x, with performance improving as GPU count increases.

Future works:

☞ **Develop a biologically realistic cerebellar circuit model** including granule layer (with Golgi cells), Purkinje layer, and other interneurons.

☞ **Benchmark network performance across multiple institutions** to evaluate and understand the scalability and performance limits of HPC systems.

Acknowledgment

This work was supported by IBS-R001-D2-2023-a00